Attention-Sink Signatures Dissociate During Vision–Language Pretraining

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Abstract

Hallucination is one of the most pressing limitations of large language models, and in vision–language models it is more rampant still. No clean solution has emerged, but several measurable symptoms track it. Attention sinks are among the most studied: positions that absorb a disproportionate share of attention regardless of content, repeatedly linked to hallucination in deployed VLMs, and partially mitigated with verifiable gains. In text language models the sink is well characterised, and it carries three signatures that co-occur so reliably they are usually treated as facets of a single phenomenon: attention concentration ($\mathrm{Sink}^{\varepsilon}{}_{1}$, per query head), a near-zero value-norm at the attended token (v-ratio, per KV head), and an outsized residual-stream norm at that position (h-ratio, per layer), which we read as a massive-activation proxy rather than a channel-level measurement of massive activations. In vision–language models the picture is far less settled.

We study how those three signatures emerge in a 222M-parameter VLM: a SigLIP-B/16 encoder feeding a randomly initialized SmolLM2-135M-architecture decoder, where position 0 is the first image token and there is no BOS. We train it under four levers: standard softmax attention, output-gated softmax, unnormalized sigmoid attention, and decoder initialization from a pretrained text LM. The encoder is pretrained in every arm, so this is vision–language pretraining with randomly initialized decoders, not fully from-scratch training. Logging all three signatures separately throughout training, we find they come apart. Across n = 2–3 seeds per arm the four levers land in four distinct corners of the three-signature space, and no two arms share a signature triple; the value-norm axis alone is strongly drained, mildly drained, or amplified, and which of the three occurs differs by lever. On a low-repetition run over one billion fresh tokens (2.39 effective visual epochs, with a held-out loss that stays stable through the full billion tokens), the massive-activation proxy more than doubles while attention concentration stays at exactly zero. The per-head relationship between concentration and value-norm flips sign from one arm to the next, so no consistent-sign relationship holds across arms.

Prior text-only work separates massive activations from attention sinks. We add dense, joint tracking of all three signatures under randomly initialized decoders in the multimodal setting, and show that all three, value-norm drain included, respond separately to ordinary training-time choices. Whether that dissociation predicts grounding or hallucination behaviour we do not test here; we establish when and how each signature forms, which is the step that comes first.

1. Introduction

Decoder-only transformers learn a habit early in training. They send a large share of attention to the first token or tokens of a sequence, whatever those tokens contain. The habit has practical weight. Streaming inference methods depend on it [1]. The extreme activation outliers that come with it make quantization harder [2]. A survey now organizes a subfield around how to interpret the habit and how to remove it [3].

In text language models the sink does not arrive alone. Three measurements move together. Attention concentrates on the sink token. The value vector at that token drops to a near-zero norm, which the literature calls value-state drain [4]. The residual stream at that token grows an abnormally large norm, which the literature calls a massive activation [2, 5]. Gu et al. [6] report that these effects settle on the same few tokens. Queipo-de-Llano, Arroyo et al. [7] report that they appear early in training, near step 1000. This regular co-occurrence has led the field to treat the three signatures as facets of one attention-sink phenomenon.

The field disputes whether they really are one phenomenon. One line of work argues for causal unity. Massive activations in the residual stream mathematically require representational compression. Ablation of those activations removes both compression valleys and sink formation [7]. A related view holds that outlier-driven rescaling by attention and residual sinks is *essential* to stable transformer training [8]. Two recent text-only studies pull the other way. Each separates a pair of these signatures by intervention. A change of normalization scheme crushes the massive-activation spike, and the sink ratio survives [9]. A value-scale intervention in from-scratch text language models keeps the sink and suppresses massive activations [10]. Both results are text-only. Each one separates at most two of the three signatures.

Vision–language models make a sharp instrument for this question, for two reasons. First, the multimodal setting changes what position 0 *is*. Our sequence starts with 49 image tokens and has no BOS token. The candidate sink token is therefore a visual token. It has none of the special-token machinery that text-LM sink accounts use. Second, the multimodal sink studies we know analyze mostly frozen, already-trained backbones at inference time [11, 12]. They show that vision-side and language-side sinks have different origins. Luo et al. [11] do track sink-dimension magnitudes across alignment checkpoints, with a frozen vision transformer and a pretrained language model. A reader still cannot see, in that setting, whether the three signatures arrive together, in sequence, or independently while they form in a decoder that starts from random weights. That requires pretraining with a randomly initialized decoder and all three signatures logged separately from step 0.

The question also carries deployment stakes. Attention allocation is the mechanism that connects the language a vision–language model generates to the content of the image. A sink is a measurable failure mode of *where* that allocation goes. A growing literature ties these same signatures to hallucination in deployed vision–language models. Visual attention sinks absorb attention, massive activation of specific hidden dimensions drives them, and redistribution methods recover the absorbed attention [13]. Hallucination errors cluster in the decoding steps immediately after the model generates a sink token, which motivates sink-aware decoding [14]. One paper proposes a causal chain from positional encoding through massive activations to visual sinks and then to hallucination [15]. In text language models, sink attention together with value-norm structure carries enough signal to detect hallucinations [16]. All of that work probes or intervenes on models that are already trained. The question underneath it is when these signatures form during training, and whether they form as one thing or as several. This paper answers that question. We do not measure whether their dissociation predicts grounding or hallucination behavior. This paper establishes the training-dynamics precondition.

We work at deliberately small scale. The model is a 222M-parameter nanoVLM [17]. A pretrained SigLIP-B/16 encoder [18] feeds a randomly initialized decoder with the SmolLM2-135M architecture [19]. We train it under four levers. Each lever targets one sink-relevant mechanism. *baseline* uses standard softmax attention. *g1gate* adds a Qiu-style G1 output gate in our zero-initialized variant, an elementwise gate on attention output, which removes both the sink and massive activations in text language models [20]. *sigmoid* replaces softmax with unnormalized sigmoid attention. That removes the normalization from which sinks are argued to come [6]. *textinit* initializes the decoder from the pretrained SmolLM2 text language model, and thus imports whatever sink structure text pretraining built. A validated probe logs concentration per query head, value-norm ratio per KV head, and the residual-norm ratio per layer, every 100 steps. The four-arm comparison reuses a small image pool; we therefore re-test the central negative result under low repetition, on a fresh stream of one billion tokens (*RF*, 2.39 effective visual epochs).

Contributions.

1. **Four levers, four corners (n = 2–3 seeds/arm).** The four arms reach four different corners of the three-signature space. No two arms share a signature triple. The value-norm ratio alone moves in three qualitatively different directions, strongly drained or mildly drained or amplified, and which one occurs differs by lever (Fig. 1, Table 2). The four arms are not a factorial design, so we report distinct intervention-associated profiles, not isolated causal effects.

2. **Low-repetition decoupling at 1B tokens (n = 1).** On a fresh stream at 2.39 effective visual epochs, with a held-out loss that never turns upward, the massive-activation proxy rises from an h-ratio of 1.43 to 3.22, about 2.3×, across a full billion tokens. Attention concentration stays at exactly zero for the entire single-seed run. No head ever crosses the sink threshold (§3.2, §5).
3. **No consistent-sign head-level relationship.** The per-head correlation between concentration and value-norm flips sign across arms (+0.76 baseline → −0.79 textinit, pooled −0.20, over 90 KV groups per arm). We report these descriptively (§3.3).

Text-only work already separates massive activations from concentration through normalization [9] and value-path [10] interventions. Decoupling by itself is therefore not new, and we do not claim it. The new part is the conjunction: dense, joint tracking of all three signatures under randomly initialized decoders in a multimodal model, with value-norm drain as a third axis that moves on its own.

2. Setup

Model and token layout. All runs use a 222M-parameter nanoVLM [17]. A pretrained, trainable SigLIP-B/16 vision encoder [18] feeds a decoder with the SmolLM2-135M architecture [19] through a learned modality projector. The decoder has 30 layers of grouped-query attention. Each layer has 9 query heads that share 3 KV heads. The decoder therefore has 270 (layer, query-head) pairs but only 90 (layer, KV-group) value projections. This distinction matters when we count per-head observations (§3.3). The decoder trains **from random initialization** in every arm except *textinit*, because the point is to watch the signatures form. Each sequence holds 49 image tokens as a causal prefix, then 79 left-padded text tokens, for 128 tokens in all. **Position 0 is the first image token, and there is no BOS token.** What happens at position 0 is therefore a property of the visual prefix, and not of inherited BOS machinery, in the arms with a randomly initialized decoder. *textinit* is the exception by design: it imports first-position structure from text pretraining, and it shows a sink before any multimodal step (§3.1).

Arms. We use four training levers. Each lever targets one sink-relevant mechanism. Everything else stays byte-identical across arms.

arm	attention	LM init	ViT init	lever precedent
baseline	softmax	random	pretrained	—
g1gate	softmax + elementwise σ-gate (zero-init, post-SDPA)	random	pretrained	G1 gating [20]
sigmoid	unnormalized sigmoid, no softmax	random	pretrained	Gu et al. [6]
textinit	softmax	pretrained SmolLM2-135M	pretrained	— (novel control)

The *g1gate* lever and the *sigmoid* lever have established sink effects in text-only models [20, 6]. The *textinit* lever has no precedent in the sink literature. It works as an inheritance control. It imports whatever sink structure text pretraining already built into SmolLM2.

A scale confound in our gate variant. Qiu et al. [20] use ordinary initialization for the G1 gate. We initialize the gate parameters at exactly zero, so the sigmoid opens at σ(0) = 0.5 at step 0. Our gated arm therefore begins as a half-scale attention-output intervention as well as a gating one, and the two effects are not separated here. We call the arm **Qiu-style G1 in our zero-initialized variant** throughout, and we repeat the caveat in §5. Any comparison to the published G1 result carries it.

Data and the two training regimes. The four-arm comparison trains on four curated subsets of `the_cauldron` [21], which hold about 146K images. The arms are matched at about 100M tokens each. *textinit* stops at 60M tokens, because it reaches its three-signature floor earlier (§3.1). Reuse of a 146K-image pool gives high visual-epoch counts. A reader could therefore treat a “no sink emerges” result as an overfitting artifact. The **RF** arm (random-fresh) answers that objection. It re-trains the *baseline* recipe on a fresh FineVision stream [22] to 1B tokens. That stream holds about 4.6M natural images and gives **2.39 effective visual epochs**, against about 74 epochs for the 1B-token repeated pool. The 100M-token comparison arms sit at roughly 7 epochs, so the 74-epoch figure describes the repeated pool at 1B tokens, not those arms. RF is therefore a **low-repetition** run, not a repetition-free one: examples do repeat, about 2.4 times on average. What we observe is that its held-out loss stays stable, and falls throughout, with no upward turn (§2, recipe). A change of dataset also trades the repetition confound for a domain-shift confound. We accept that trade and document it. The fresh pool holds natural images and leans heavily on COCO. We estimate its overlap with the repeated subsets at under 3% from the config-level composition of the two pools; we did not run image-level deduplication, so that number is an estimate, not measured evidence. We did not run a third, domain-matched control (§5).

Three signatures, tracked separately. We log the three sink symptoms that the text-LM literature reports together, each at its own granularity, following Gu et al. [6] at a fixed sequence length. The decoder has $L = 30$ layers, $H = 9$ query heads per layer, and $G = 3$ KV heads per layer. Every quantity below is averaged over valid query positions and over the fixed probe batch.

Concentration ($\text{Sink}^{\varepsilon}_1$) is the fraction of the $L{\cdot}H = 270$ (layer, query-head) pairs whose mean attention to position 0 exceeds ε. We use the $\varepsilon = 0.3$ default of [6] and check $\varepsilon \in \{0.2, 0.4\}$. Cross-arm tables report the stricter $\varepsilon = 0.2$, which makes an absence claim harder to pass.

Value-norm ratio (v-ratio) is the value norm at position 0 divided by the mean value norm over the other valid positions, computed per layer and then averaged over the 30 layers. Below 1 is value-drain [4]; above 1 is amplification. Under grouped-query attention a value vector belongs to a KV group and repeats across 3 query heads, so the ratio rests on $L{\cdot}G = 90$ independent value projections, not 270. This matters for §3.3.

Residual-norm ratio (h-ratio) is the residual-stream norm at position 0 divided by the mean norm over the other positions, again per layer and then averaged over the 30 layers. We call it a **massive-activation proxy**. Massive activations are normally defined by channel-level outliers [2, 5], which we never measured; the h-ratio captures a position-specific residual-norm asymmetry, necessary but not sufficient for that definition.

Why a sink is expected at all. Softmax forces every attention row to sum to one. A head with nothing informative to retrieve must still place its mass somewhere. Gu et al. [6] argue that the sink token absorbs that surplus and acts "more like key biases." That sum-to-one constraint is the standing mechanism against which our *sigmoid* arm, which removes normalization, is the direct test.

All three metrics **anchor on position 0 by construction**. We state this as a measurement choice, and we check it. At seed 0, per-position attention mass makes position 0 the maximum-mass token in every arm (Appendix C). Section 5 handles the remaining seed-level caveat. The *sigmoid* arm reports the row-normalized attention view, which keeps concentration comparable across arms. We also log the raw sigmoid mass.

Probe. The function `probe_sinks()` re-walks the decoder from the live module weights in eager mode, in fp32, with no gradients. It is independent of the training path, which uses fused SDPA kernels, autocast, and `torch.compile`. Every call validates its hidden states against the real forward pass of the model, to a relative error below 10^{-2}. The probe therefore cannot drift from what the trained model computes. The probe batch stays fixed across all runs and all seeds, so every number in this paper is comparable from run to run. Probes fire every 100 optimizer steps. That cadence is dense enough to timestamp the first threshold crossing of each signature (§3.4).

Recipe. We use AdamW with weight decay 0.1, following [6], and a gradient clip of 1.0. The schedule is cosine with 3% warmup. The learning rates are 4×10^{-4} for the language model, 2×10^{-3} for the projector, and 10^{-4} for the vision encoder. We use bf16 autocast, `torch.compile`, and a batch size of 128. Arms in a comparison differ only in the lever under test. Training stays healthy in all reported runs.

Reporting details. *Token accounting:* one token is one image token or one non-padding text token. A full sequence contributes 49 image tokens plus its non-padding text tokens, so a step of batch 128 contributes at most 16,384 tokens and about 9.5K on average. All "M tokens" figures in this paper use that definition. *Probe:* the fixed probe batch holds n = 32 samples, drawn with `random.seed(0)` from the repeated `the_cauldron` tail; we label this probe version `v1–repeatedtail–32`. RF uses the same probe batch as the repeated arms, so its signatures stay comparable to theirs even though its training stream differs (§5). Probes fire every 100 steps. *Validation:* a 1,024-example held-out pool, evaluated every 500 steps; each estimate is the mean over the first 512 examples of that pool (four batches of 128, in fixed order). We report `val_unseen`, which holds out images. For the repeated arms we also log `val_seen`, which re-uses training images with fresh question–answer text. The RF run has no distinct seen split: at 2.39 visual epochs its `val_seen` loader re-uses the held-out pool, so we report only `val_unseen` for RF. *Seeds:* two for *baseline* and *sigmoid*, three for *g1gate* and *textinit*, one for *RF*. *Aggregation:* the three formulas above; each per-layer or per-head quantity is first averaged over the probe batch and over valid query positions, then aggregated as written.

Validation losses, and what they do and do not license. At the matched 100M-token checkpoint the held-out losses are 1.182 for *baseline*, 1.133 for *g1gate*, 1.206 for *sigmoid*, and 0.877 for *textinit* (seed 1 or 2; seed-0 archive values are 1.161, 1.138, 1.108, and 0.832). RF reaches 0.638 at 1B tokens. Two cautions. First, *textinit* starts from a pretrained text decoder, so its lower loss reflects unequal competence, not a lever effect; only *baseline*, *g1gate*, and *sigmoid* are equal-token, equal-initialization comparisons. Second, the repeated-data arms show a large train–validation asymmetry (`val_seen` near 0.44 against `val_unseen` near 1.18), which is the overfitting signal that motivates RF. RF cannot be checked the same way, because it has no distinct seen split. The weaker statement its data supports is that the held-out loss falls throughout and never turns upward: it goes from 1.35 over the first ten evaluations to 0.68 over the last ten, and the fitted slope over the second half of the run stays negative. **We did not run MMStar or any other downstream benchmark on any arm**, so this paper makes no capability claim (§5).

3. Results

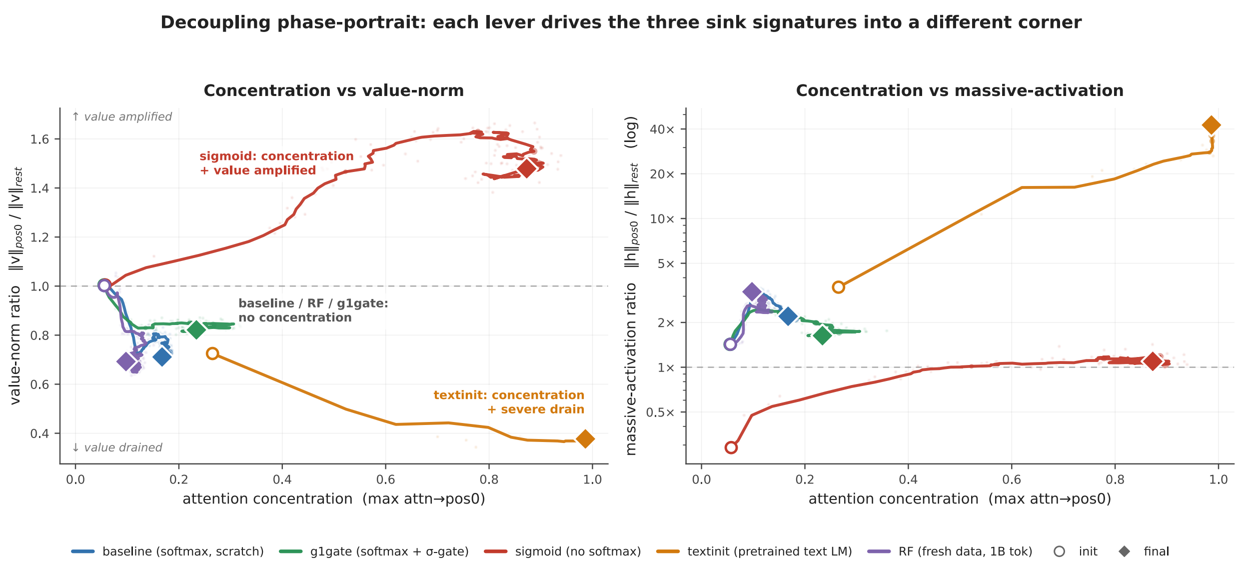

Figure 1: Decoupling phase-portrait. Training trajectories of every arm in concentration-against-value-norm space, at left, and concentration against the residual-norm ratio, at right, on a log scale. **Note the metric difference from the tables:** the horizontal axis here is the *maximum* attention→pos0 over heads, a continuous quantity that keeps a trajectory visible, whereas Tables 1–2 report $\mathrm{Sink}^{\varepsilon}{}_{1}$, the *fraction of query heads above a threshold*. An arm can therefore show a non-zero position on this axis while its $\mathrm{Sink}^{\varepsilon}$ remains 0.000. The value-norm ratio is per KV head and the h-ratio is per layer (§2). Paths are smoothed with a moving average, 5 points on the horizontal axis and 9 on the vertical; the faint dots behind each path are the unsmoothed per-probe values, and the init and final markers use raw values. Circles mark initialization. Diamonds mark the final checkpoint, at about 100M tokens for *baseline*, *g1gate* and *sigmoid*, 60M for *textinit*, and 1B for *RF*; seed 0 unless a seed is labelled. Each lever drives the signatures to a different corner. *sigmoid* (red) reaches strong concentration, *amplifies* the value norms, and shows no strong residual-norm asymmetry. *textinit* (orange) combines strong concentration with severe value-drain and an extreme residual-norm ratio. *baseline*, *g1gate*, and *RF* (blue, green, violet) never leave the no-concentration region, and their residual-norm ratio grows. If the three signatures were one phenomenon, these trajectories would move together along one direction. Instead every combination of directions occurs.

3.1 Four training levers reach four different signature corners

Table 1 reports the three signatures for all four arms and all seeds. *baseline* and *sigmoid* have two seeds each. *g1gate* and *textinit* have three. Each row is read at its final matched checkpoint: about 100M tokens for baseline, g1gate, and sigmoid, and 60M tokens for *textinit*. All three signatures of *textinit* have plateaued at 60M tokens. Its h-ratio changes by less than 10% from 60M to 100M in the seeds that continued. We trust the seed-0 values from a check-summed archive rather than re-derive them first-hand (§5).

Table 1 — signature triple by arm, all seeds. $\mathrm{Sink}^{0.2}{}_1$ is the fraction of the model's 270 (layer, query-head) pairs whose mean attention→pos0 exceeds 0.2. The v-ratio aggregates 90 (layer, KV-head) value projections and the h-ratio aggregates 30 layers (§2).

arm	seed	tokens	$\mathrm{Sink}^{0.2}{}_1$	mean attn→pos0	v-ratio	h-ratio
baseline	s0	101M	0.000	0.068	0.723	2.16
baseline	s1	100M	0.000	0.062	0.687	1.71
g1gate	s0	101M	0.004	0.068	0.805	1.67
g1gate	s1	100M	0.011	0.073	0.845	2.04
g1gate	s2	100M	0.0037	0.072	0.854	2.24
sigmoid	s0	102M	0.830	0.377	1.479	1.10
sigmoid	s1	100M	0.756	0.311	1.603	1.30
textinit	s0	60M	0.852	0.627	0.377	42.5
textinit	s1	60M	0.556	0.235	0.634	5.50
textinit	s2	60M	0.578	0.232	0.484	12.2

Table 2 collapses the seeds into per-arm ranges. **No two arms share a signature triple.** The value-norm axis alone takes three qualitatively different directions. The lever drains it hard below 1, drains it only mildly, or amplifies it above 1. No arm leaves it unchanged.

Table 2 — the four corners.

arm	concentration	value-norm	massive-activation proxy
baseline	absent (0.000)	mild drain (0.69–0.72)	moderate (1.7–2.2)
g1gate	near-absent (0.004–0.011)	**milder drain** (0.81–0.85)	moderate (1.7–2.2)
sigmoid	strong (0.76–0.83)	**amplified** (1.48–1.60)	no strong asymmetry (1.1–1.3)
textinit	strong (0.56–0.85)	strong drain (0.38–0.63)	extreme (5.5–42.5)

The corners separate pairwise on single axes. *g1gate* differs from *baseline* on the value-norm axis alone. Concentration is absent or near-absent in both arms (note that *g1gate* sits *above* the exact 0.000 of baseline, not below it), and the residual-norm ratio is moderate in both. The gate nonetheless makes the mild value-drain of the baseline milder still, from 0.69–0.72 to 0.81–0.85, which is a 15–19% drain rather than none. *sigmoid* and *textinit* both reach strong concentration, and they then part on two axes. They differ in the *direction* of the value-norm move, amplified against drained, and they differ in the residual-norm ratio, absent against extreme. Each arm shows a distinct intervention-associated profile. Because the four arms are not a factorial design, we do not attribute these profiles to isolated causal effects of single levers. Figure 1 shows the four trajectories separating early and ending in four different corners. They do not share one origin: *textinit* starts from a pretrained text decoder and therefore begins with an inherited sink already in place (§3.1).

Reproducibility differs by arm. *g1gate* is the tightest arm. Its $\mathrm{Sink}^{0.2}{}_1$ values are 0.004, 0.011, and 0.0037 across three seeds, which is at most 1% of heads. *baseline* and *sigmoid* are tight in both of their seeds. The corner of *textinit* repeats across seeds. Its magnitudes do not. All three textinit seeds show high concentration, strong value-drain, and an h-ratio above 5. Seed 0 nonetheless sits far above the other two seeds on every signature at once. Sink reads 0.85 against 0.56–0.58, mean attn→pos0 reads 0.63 against 0.23, and h-ratio reads 42.5 against 5.5–12.2. The h-ratio has plateaued by 60–100M tokens in both lower seeds. The spread is therefore genuine seed sensitivity, not an unconverged transient. Part of it is also positional: at seeds 1 and 2 the arm's peak signatures move off position 0, where our metrics are anchored (§3.4). We report the massive-activation proxy of textinit as a **range (5.5–42.5×) with a median near 12×** everywhere in this paper. We treat the corner as the reproducible claim, and no particular magnitude.

3.2 The massive-activation proxy grows across 1B low-repetition tokens while concentration stays at zero

The four-arm comparison reuses a 146K-image pool. A reader could therefore read its "concentration never emerges" result as an overfitting artifact. The RF arm addresses that. It runs the identical *baseline* recipe on a fresh FineVision stream to one billion tokens, at **2.39 effective visual epochs**. Examples still repeat, about 2.4 times on average, so this is a low-repetition run rather than a repetition-free one. Its held-out loss falls through the full billion tokens and never turns upward, so nothing in the run indicates overfitting. RF has no distinct seen split, so we cannot repeat for it the `val_seen` / `val_unseen` comparison that exposes memorization in the repeated arms (§2, §5).

Concentration never arrives. **$\text{Sink}^{0.3}{}_1$ = 0.000 across the entire 0 → 1B run.** No head of the 270 crosses the sink threshold, at any of about 700 probes. The massive-activation proxy, in the same run, keeps growing far past warmup.

Table 3 — RF (fresh stream, 2.39 visual epochs, seed 0), init → 1B tokens.

signature	@ init	@ 1B	net
$\text{Sink}^{0.3}{}_1$ (concentration)	0.000	0.000	flat zero, entire run
max attn→pos0	0.056	0.098	≈flat, far below threshold
h-ratio (massive-activation proxy)	1.43	**3.22**	**≈2.3×**, positive long-horizon trend
v-ratio (value-norm)	1.00	0.69	net drain, non-monotone

Warmup does not explain the rise in h-ratio. The h-ratio climbs from 2.40 at about 57M tokens to 3.22 at 1B tokens, long after the 3% warmup window closes. The rise is not monotone at probe resolution; we report it as a positive long-horizon trend. In Figure 1, at right, the violet trajectory climbs and never moves rightward. The v-ratio ends below its value at initialization. About 75% of that drop happens in the first 57M tokens, and the v-ratio then recovers part of it. We therefore report the v-ratio as supporting context, not as ongoing emergence. **The massive-activation proxy grows about 2.3× across a full billion tokens of multimodal training while attention concentration never leaves zero.** This run uses a single seed (§5).

3.3 No consistent-sign head-level relationship between concentration and value-norm

A tight coupling between concentration and value-drain should at minimum hold the sign of their per-head correlation constant across training regimes. It does not. Table 4 reports the Pearson r between per-head attention→pos0 and value-norm ratio. We read it at the final checkpoint of each arm, over the 90 (layer, KV-group) observations per arm: under grouped-query attention a value vector is shared by 3 query heads, so we average the attention of a group's query heads rather than triplicate its value observation (§2). Appendix Fig. A2 plots the scatter.

Table 4 — per-head r(attn→pos0, v-ratio), final checkpoint. Descriptive only; we report no p-values, for the reason given below.

baseline	g1gate	sigmoid	textinit	RF	pooled
+0.76	+0.57	−0.04	−0.79	+0.51	−0.20

These correlations are **descriptive statistics, not inferential ones**, and we deliberately report no significance tests. Collapsing to the 90 KV groups removes the pseudoreplication that a per-query-head reading would introduce, and it leaves the picture unchanged: at 270 pairs the same quantities read +0.67, +0.53, −0.03, −0.76 and +0.43, so every sign and every ordering survives. Heads within a layer are still not independent, however, and each arm rests on a single seed at this checkpoint, so a p-value would remain misleading.

The pattern we report is about **sign**, which the pseudoreplication does not manufacture. Heads that attend more to position 0 have *larger* value norms in the baseline regime and *smaller* value norms under text initialization. The pooled correlation is weak only because arms of opposite sign cancel; it is not an uncorrelated cloud. We therefore claim no consistent-sign relationship across arms, and we do not claim the absence of a coupling law: a fixed coupling would have to show one sign everywhere, and it does not.

3.4 Ordering: norm signatures lead, and concentration is late, mirrored, or absent

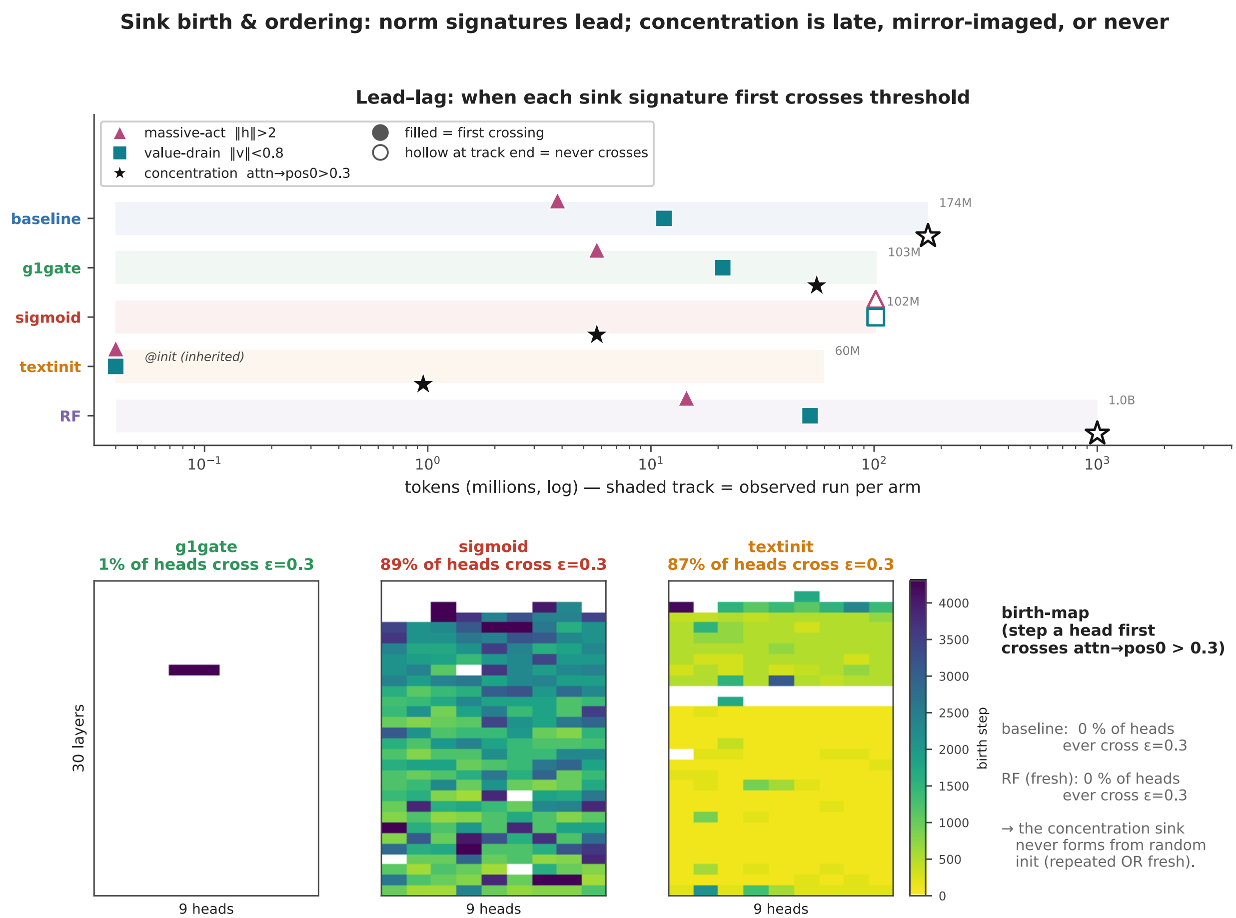

Figure 2: Sink birth and ordering. *Top:* a time-to-event view. Each shaded track spans the tokens for which we observed that arm, from 60M to 1B. A filled marker shows the first probe at which a signature crossed its per-head threshold: massive activation h>2, value-drain v<0.8, concentration attn→pos0>0.3. A hollow marker at the end of a track means that the signature never crossed inside the observed run. In the softmax arms with random decoders, baseline and g1gate and RF, the norm signatures cross within about 4–50M tokens. Concentration in those arms never crosses, through 174M tokens for baseline and through 1B tokens for RF, or barely crosses, in g1gate. *sigmoid* mirrors that pattern. Its concentration crosses at about 6M tokens, and neither norm signature ever crosses. *textinit* inherits its norm signatures at 0 tokens and crosses concentration before 1M tokens. *Bottom:* birth-maps, which show the step at which each (layer, query-head) pair first crosses the concentration threshold. No baseline head and no RF head ever crosses. 1% of g1gate heads cross, against 89% for sigmoid and 87% for textinit. **All crossing times are interval-censored** at the probe cadence: a signature is observed only every 100 optimizer steps, so a reported birth step means "the first probe at which the threshold had already been crossed", and the true crossing lies somewhere inside the preceding interval. Two signatures that cross within one interval of each other should not be read as ordered. Seed 0 throughout.

Dense probing lets us timestamp the arrival of each signature (Figure 2). The ordering stays consistent inside each arm family. In the softmax arms with random decoders, *baseline* and *g1gate* and *RF*, the norm signatures come early. Massive activation crosses its per-head threshold within the first few to about 15M tokens, and value-drain follows within about 50M tokens. Concentration never comes. No baseline head and no RF head crosses the concentration threshold at any point in training. g1gate reaches 1% of heads, a single-layer blip late in training. *sigmoid* mirrors that pattern. Its concentration crosses early, at about 6M tokens, and 89% of its heads cross eventually, while neither norm signature ever crosses. *textinit* inherits its norm signatures rather than grows them. Value-drain and an elevated residual-norm ratio are both present at 0 tokens, imported with the pretrained text-LM weights. Concentration is *not* inherited in the same way: $\mathrm{Sink}^{0.3}{}_1$ is 0.000 at step 0 and crosses before 1M tokens. Even in the arm that imports the most sink structure, the norm signatures precede concentration.

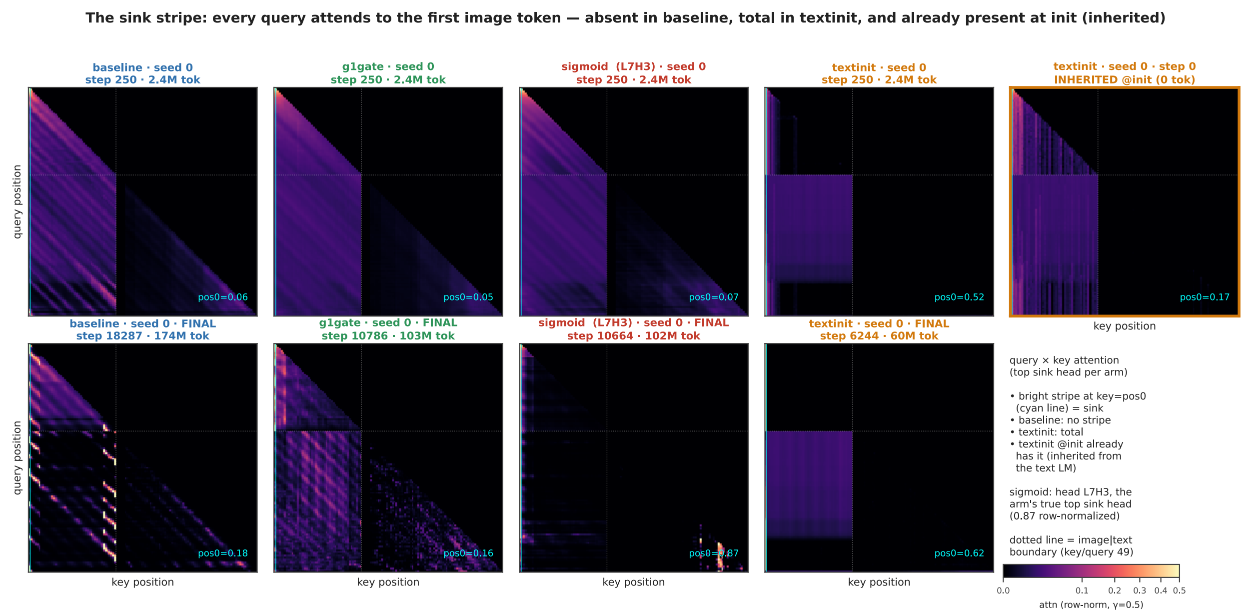

Figure 3: The sink stripe. Query × key attention of the top sink head of each arm, row-normalized, at seed 0; every panel is labelled with its seed, step and token count. The top row is early in training (step 250, about 2.4M tokens). The bottom row is the final checkpoint. The cyan line marks key = pos0. The dotted line marks the image-to-text boundary. The pos0 stripe is *absent* in baseline throughout, and strong in textinit, which reaches attn→pos0 = 0.62 at its final checkpoint. The *sigmoid* column uses head **L7H3** (0-indexed), the arm's top sink head on the paper's fixed probe batch at 0.87 row-normalized attention to pos0; an earlier dump selected heads by raw, unnormalized gate score and picked different heads, which understated the arm. The sigmoid heat maps are re-rendered from an auxiliary streaming batch, because the saved matrices covered the superseded heads; the printed pos0 shares for that column, and every sigmoid number in the text and tables, come from the fixed probe batch. The rightmost panel shows the stripe *already present at step 0* in textinit, inherited from the text language model. The pos0 share printed on each panel is computed over valid query positions, the same convention as the tables.

Figure 3 shows the same result at the level of a single head. It plots the query × key attention map of the top sink head of each arm. The vertical stripe at key = pos0 marks every query attending to the first image token. That stripe is absent in *baseline* at both the early and the final checkpoint, and strong in *textinit*. The rightmost panel carries the most weight. **The stripe is already there at step 0 in textinit**, imported with the text-LM weights before the model has seen one image.

Raw against row-normalized sigmoid. The *sigmoid* panel needs one measurement note, because row-normalization changes the object being measured and Gu et al. [6] state their result for *unnormalized* sigmoid attention. Head L7H3 of the sigmoid arm sends 0.873 of its row-normalized attention to position 0 at the final checkpoint, which is the arm maximum. Its raw gate mass to position 0 is 0.065, and the raw mass summed over all keys in that row is 0.110: the row does not sum to one, and pos0 takes about 59% of what little gate mass the head opens at all. Early in training the same head shows the opposite picture: raw pos0 mass 0.309 against a raw row sum of 6.44, so under 5%

of a very large gate budget. The concentration this arm develops is therefore a **relative** reallocation of a shrinking gate budget onto position 0, not the growth of a large absolute mass there. Raw and row-normalized values here come from the same fixed probe batch as every other number in this paper, masked to valid query positions. We report the row-normalized view in the tables so the arms stay comparable, and we flag the raw view here because it is the quantity Gu's claim concerns.

Attention-entropy collapse, which the text-LM literature uses as the usual correlate of concentration, separates the same way. Only *sigmoid* and *textinit* collapse (Appendix Fig. A3). Entropy collapse tracks the concentration axis, not the norm axes.

The signatures also dissociate in position. A per-position scan of attention mass, residual norms, and value norms across all three *textinit* seeds shows that the three signatures need not share a token. At seed 0 they coincide: attention mass, the residual-norm peak, and the value-norm minimum all sit at position 0. At seed 1 the attention maximum stays at position 0 while the residual peak moves to pos1 and the value minimum to pos5. At seed 2 they separate furthest: the attention maximum sits at **pos1** while the residual peak and the value trough both sit at **pos13**. The softmax arms with randomly initialized decoders behave differently: position 0 remains the maximum-mass token in *baseline*, *g1gate*, and *sigmoid* at every seed we scanned (Appendix C). In *RF* the attention argmax sits at pos1, but the mass there is 0.100 against 0.083 at position 0, which is a diffuse profile in the baseline range rather than a concentration sink.

We report this as a supporting observation, not as a second headline. It carries one consequence for measurement: because all three of our metrics anchor on position 0 by construction, the *textinit* magnitudes at seeds 1 and 2 are read at a token that is no longer the peak, so those pos0-anchored values **understate** the arm's true peak magnitudes. That is a further reason to report *textinit* as a range and a median (§3.1, §5). The pos0 anchoring holds for the arms with randomly initialized decoders, which carry the central negative result.

3.5 Interpretation (hypotheses)

This subsection is **speculative**. Nothing in it is tested by our experiments, and none of it is a claim of this paper. We offer it because a reader is entitled to ask *why* the signatures come apart, and because these hypotheses are cheap to state and testable later.

Gu et al. [6] account for the sink as a key bias: softmax must distribute a full unit of attention mass per row, and a head with nothing informative to retrieve parks the surplus on a token whose value contributes little. If that account is right, each of our levers touches a different part of it, which would explain why each moves a different axis.

- *g1gate.* An output gate can suppress whatever the attended position injects into the residual stream. A model with that gate may be able to *afford* concentration, because concentration no longer forces a matching change in the value path. That would be consistent with a gate whose measured effect here is on the value-norm axis rather than the concentration axis. Fesser et al. [23] make a compatible argument from another direction: they distinguish a sink that acts as an "adaptive nop" (a no-op: the head suppresses its own update by routing attention to a token whose value contributes nothing), recognizable by a negligible value norm, from a sink that broadcasts global information, and they note that gating implicitly assumes the nop mechanism. If that is right, a gate should act on the value-norm axis first, which is where our gated arm differs from baseline.
- *sigmoid.* Removing the softmax removes the sum-to-one constraint itself. A head with nothing to retrieve can simply attend weakly to everything, with no surplus to park. On this reading the value amplification we observe is what the value path does when it is no longer compensating for a forced allocation.
- *textinit.* A pretrained text decoder arrives with sink machinery already built. Our observation that its norm signatures are present at step 0 while concentration crosses later would then reflect import of structure followed by re-targeting onto a visual prefix, rather than formation from scratch.

Each of these is a hypothesis about mechanism. Testing them needs interventions we did not run: gate-scale sweeps that separate gating from the half-scale confound of §2, sigmoid runs at matched effective attention temperature, and text-initialized runs with the sink machinery ablated before alignment.

4. Related Work

Attention sinks and their companions in text language models. Gu et al. [6] give the canonical empirical account. The sink token acts "more like key biases, storing extra attention scores, which could be non-informative and not contribute to the value computation." They report small key and value norms at the sink as part of the same phenomenon, and they connect the sink to massive residual-stream activations [2, 5]. They also show that unnormalized sigmoid attention prevents sink formation in text language models up to 1B parameters. Our *sigmoid* arm builds on that result. Guo et al. [4] describe the coupling between concentration and value-drain as "active-dormant" heads, where the model actively drives the value state of the sink head toward zero. Queipo-de-Llano, Arroyo et al. [7] make the strongest unity claim, and the only genuinely *causal* one. Massive activations mathematically require representational compression, and ablation of the layer-0 massive activation of a model removes both compression valleys and sink formation. Across Pythia checkpoints they find that sinks, compression valleys, and massive activations appear together near step 1000 and then stay synchronized. We reserve the term "causally unified" for that result alone. Peng et al. [24] trace a first-position sink circuit that emerges early in from-scratch text pretraining. That work covers from-scratch emergence dynamics, in text only, and does not separate the signatures.

Prior decoupling results: text-only, two axes. Two recent papers already separate pairs of these signatures in text models, and we scope our claim around them. Sun, Canziani, LeCun and Zhu [9] show that massive activations and attention sinks are dissociable architectural artifacts. A change of normalization scheme crushes the massive-activation spike, and the sink ratio survives. That is a two-way dissociation, in text only, through a normalization lever, on trained checkpoints. Chen and Yao [10] decouple the same pair from the opposite direction and from scratch. In text language models of 0.1–0.3B parameters, probed at dense checkpoints, a value-scale intervention keeps the sinks and suppresses massive activations. Neither paper treats value-norm drain as a third axis that moves on its own. Neither paper is multimodal. Fesser et al. [23] come at the same question from another direction, and their result is the closest external support for tracking the value norm separately: they argue that one sink pattern can hide two different algorithms, an "adaptive nop" (a no-op: the head suppresses its own update by routing to a null token) and a "broadcast" that aggregates and redistributes global information, and that the two leave different traces — nop sinks show negligible value norms, broadcast sinks produce low-rank outputs. On their account gating implicitly assumes the nop mechanism and registers implicitly assume broadcast. That work is on vision transformers rather than vision–language models, and it diagnoses trained models rather than tracking emergence, but it independently motivates treating the value norm as diagnostic rather than decorative. On the opposite side of the argument, Qiu et al. [8] hold that outlier-driven rescaling by attention and residual sinks is essential to stable training. The field is unsettled on two counts. It disputes whether these signatures separate, and it disputes whether anyone should remove them at all.

Multimodal sinks: mostly frozen backbones, inference time. Luo et al. [11] identify high-norm attention-sink tokens that originate in the vision transformer, and they separate ViT-propagated sinks from LLM-emerged sinks. Their Appendix A.4 does track sink-dimension magnitudes across alignment checkpoints, so a training-time view of multimodal sinks is not without precedent; that view uses a frozen vision transformer and a pretrained language model, and follows sink-dimension magnitude rather than the three signatures separately. Choi et al. [12] likewise distinguish vision-sinks from language-sinks in a frozen large vision–language model and gate them by layer. Both papers establish that multimodal sinks have distinct vision-side and language-side origins, and our *textinit* inheritance result fits that frame naturally. What is not yet available in that literature is dense, joint tracking of concentration, value-norm and residual-norm as separate quantities in a decoder that starts from random weights. Vision transformers also grow high-norm "register" tokens of their own [25], which is why the pretrained encoder gets its own limitation in §5.

Sinks and hallucination in deployed vision–language models. A parallel line of work ties these signatures to grounding failure. Kang et al. [13] attribute *visual* attention sinks to massive activation of specific hidden dimensions, and they redistribute the absorbed attention to reduce hallucination. Shukla and Kira [14] report that hallucination errors concentrate within a few decoding steps of the generation of a sink token. They then decode in a sink-aware way. Zhang et al. [15] trace a causal chain from rotary position encoding through massive activations to visual sinks and then to hallucination. In text language models, Binkowski et al. [16] detect hallucinations from sink structure. Their classifier relies preferentially on sinks whose value vectors have large norms, which makes it the closest precedent for treating value norms as a signal. All four papers work on models that are already trained, at inference time. Kang et al. [13] and Zhang et al. [15] both treat massive activation and concentration as one coupled mechanism; Shukla and Kira [14] work at the level of sink tokens and grounding rather than massive activations, and [16] is text-only as well. None of them tracks the signatures across training. None of them treats value-norm drain as a third axis that moves on its own. Our result is the training-dynamics complement to that line. The coupling on which the line depends is lever-dependent, not fixed.

The gating lever. Qiu et al. [20] introduce the head-specific elementwise sigmoid gate on attention output that our *g1gate* arm adapts. In text language models the gate "largely reduces the attention score allocated to the first token and decreases massive activations" while it improves quality. Our variant differs in one way that matters: Qiu et al. use ordinary initialization, and we initialize the gate at exactly zero, so it opens at $\sigma(0) = 0.5$ and applies a half-scale factor to attention output at step 0 (§2). Our arm is therefore **Qiu-style G1 in a zero-initialized variant**, and gating and initial scaling are confounded in it. With that caveat, what we observe in the multimodal setting with a randomly initialized decoder is that concentration is already absent in the baseline, so the axis on which our gated arm differs from baseline is the value-norm axis: the drain becomes milder, from a v-ratio of 0.69–0.72 to 0.81–0.85, rather than being removed. The residual-norm ratio is comparable in the two arms. That difference stays invisible unless the three signatures are logged separately.

Positioning. The closest prior work separates at most two of the three axes, in text-only models, through normalization or value-path interventions. The multimodal studies work mostly on frozen models, and where one tracks alignment checkpoints [11] it follows sink-dimension magnitude rather than the three signatures separately. Our claim is therefore the conjunction: **dense, joint tracking of concentration, value-norm drain, and the residual-norm ratio as separately measured quantities, in multimodal pretraining with a randomly initialized decoder**, which adds value-norm drain as a third axis beyond the massive-activation-vs-sink dissociations shown in text models [9, 10]. We do not claim priority on decoupling itself, and we do not claim that prior multimodal work cannot study emergence.

5. Limitations

Metrics anchored on position 0. We measure all three signatures at the first image token. We verified this anchoring through per-position attention *mass*, not through argmax alone. At seed 0 position 0 is the maximum-mass token in every arm: the mean and max-head values are 0.06 and 0.17 for baseline, 0.07 and 0.23 for g1gate, and 0.30 and 0.66 for sigmoid, and 0.63 and 0.99 for textinit, whose next-highest position reads 0.009 (Appendix C). We then scanned attention mass, residual norms and value norms per position at the other seeds. Position 0 stays the maximum-mass token for *baseline*, *g1gate* and *sigmoid* at every seed scanned. It does **not** in *textinit*: at seed 1 the residual peak moves to pos1 and the value minimum to pos5 while attention stays at pos0, and at seed 2 the attention maximum sits at pos1 while the residual peak and value trough sit at pos13 (§3.4). Our pos0-anchored magnitudes for textinit at those seeds therefore *understate* the arm's peak values, which is a further reason we report textinit as a range and a median and treat its corner rather than its magnitudes as the claim. In *RF* the attention argmax sits at pos1, but with mass 0.100 against 0.083 at pos0, a diffuse profile rather than a displaced sink; the RF negative result does not depend on the anchor.

The gate arm carries a scale confound. We initialize the G1 gate at exactly zero, so it opens at $\sigma(0) = 0.5$ and halves attention output at step 0, where Qiu et al. [20] use ordinary initialization (§2, §4). Gating and initial output scaling are therefore confounded in our *g1gate* arm, and its differences from baseline cannot be attributed to gating alone. A scale-matched control is future work.

Pretrained vision encoder: no arm is fully from scratch. The SigLIP encoder is pretrained and trainable in every arm, and *textinit* additionally uses a pretrained decoder. What we study is therefore vision–language pretraining with randomly initialized decoders, not from-scratch training of the whole model, and we word it that way throughout. Vision transformers grow high-norm register tokens of their own [25], and sinks can propagate from a vision transformer into a large vision–language model [11]. Part of our residual-norm signal could therefore be inherited rather than decoder-formed. Our defense is the trajectory. The h-ratio starts at 1.0–1.4 at initialization and *rises* through training, from 1.43 to 3.22 across 1B tokens in the RF arm. Pure inheritance from a static encoder predicts a high, flat h-ratio from step 0 instead. A control with a randomly initialized vision transformer, which would isolate the decoder entirely, is future work.

The h-ratio is a proxy, not a measurement of massive activations. Massive activations are normally defined by channel-level outliers [2, 5]. We measured a position-specific residual-norm ratio and never computed channel-level statistics, so we report the h-ratio as a massive-activation proxy (§2). A large h-ratio is consistent with massive activations but does not establish them.

Token scale. Our runs reach at most 1B tokens per arm, against the roughly 5B tokens that are canonical in the text-LM sink literature [6]. Sink emergence is early relative to that budget. Text-LM sinks and their companions form near step 1000 [7], far inside our range. We nonetheless cannot rule out that a signature absent at 1B tokens emerges later. Confirmation at larger scale is future work.

Reproducibility of textinit magnitudes. The massive activation of *textinit* is seed-sensitive. The h-ratio spans 5.5–42.5 across three seeds, and seed 0 is the consistent outlier on every signature. The corner is the reproducible claim: high concentration, plus strong drain, plus large massive activation. No specific magnitude is.

Provenance and seed count. We trust the seed-0 raw probes for the four-arm comparison from a checksummed archive summary rather than re-derive them first-hand. We *did* independently re-derive seeds 1 and 2. That audit caught and corrected a metric-labeling error in an earlier internal consolidation, which mixed mean and max attention in one column. That correction is why we report the metrics like-for-like here. *baseline* and *sigmoid* have two seeds. *g1gate* and *textinit* have three. The RF arm uses a **single seed**, and it contains one **weights-only optimizer restart at about 57M tokens** that an out-of-memory error forced: the model weights were reloaded and the AdamW moment estimates were discarded, so RF is not a single uninterrupted optimizer trajectory. The audit verified signature continuity across that seam. The v-ratio and h-ratio values are identical at the shared checkpoint. A double-covered 600-step overlap diverges only within probe noise, and concentration reads 0.000 on both sides. The decoupling movement also completes before the seam. Concentration was reproducibly zero across both seeds of the repeated-data baseline, which we take as adequate support for the negative claim. A second fresh-data seed would strengthen it.

Stream-order and probe-batch caveats on RF. Two further details of the RF run are worth stating. The streaming shuffle buffer was reduced from 1500 to 500 examples partway through, again for memory reasons, so stream ordering is not homogeneous across the run; the $\mathrm{Sink}^{0.3}{}_1 = 0.000$ result and the h-ratio rise both hold within each regime separately. And the RF probe batch is still the **fixed repeated-`the_cauldron` tail** used by the other arms, not a sample of the FineVision stream. That choice keeps RF's signatures comparable to the repeated arms, which is what the negative result needs, but it means RF's signatures are measured on data from the other distribution.

Domain shift in the fresh-data control. The RF arm reduces repetition by a change of dataset, dropping the visual-epoch count from about 74 — a figure that describes the 1B-token repeated pool, not the 100M-token comparison arms, which sit near 7 epochs — to 2.39. That change introduces a domain-shift confound in its place. We accepted the trade deliberately. Repetition is the dominant confound, and we chose the fresh pool to minimize shift, as natural images, COCO-heavy. We estimate the overlap with the repeated subsets at under 3% from the config-level composition of the two pools; we did not run image-level deduplication, so that figure is an estimate and not measured evidence. A domain-matched control that compares fresh and repeated data is the known follow-up.

We report no benchmark accuracy. We measure sink signatures with the probe of §2. We did not run downstream benchmark evaluation, such as MMStar, on any arm. We make no claim about how signature dissociation relates to downstream capability.

6. Conclusion

We tracked the three signatures that the field bundles as "the attention sink" as separate quantities across multimodal pretraining with randomly initialized decoders. Those signatures are concentration, value-norm drain, and a residual-norm ratio that we read as a massive-activation proxy. They came apart everywhere we looked. Four training levers produced four different signature corners, and the value-norm axis alone moved in three different directions. On a low-repetition, single-seed run over a billion fresh tokens, at 2.39 effective visual epochs and with a held-out loss that never turned upward, the massive-activation proxy grew about 2.3× while concentration never left zero. The per-head correlation between concentration and value-norm flipped sign across arms. The signatures even arrived in different orders. Norms came first and concentration never came in the softmax arms with random decoders. Sigmoid attention mirrored that order. Text initialization imported the norm signatures at step 0 and crossed concentration shortly after. In that arm the signatures separated in position as well, sitting on different tokens at two of three seeds.

The signatures are plainly related in general. Work on text language models documents real interactions among them, which include a causal route from massive activations to sinks and compression valleys [7]. Our results show that the coupling is not obligatory. In multimodal pretraining with a randomly initialized decoder, each axis moved separately under ordinary training-time levers. That extends the two-way text-only dissociations [9, 10] to a third axis and to a new setting. For interpretability work and mitigation work the practical result is blunt. One signature is not a proxy for the others. A model with no attention sink can still carry a growing residual-norm asymmetry. A gate that changes value-drain can leave the other axes where they were.

Next steps. Three steps follow from this work. Train a randomly initialized vision encoder, to isolate the contribution of the decoder to the residual-norm signal. Extend the fresh-data run past 1B tokens, to match text-LM budgets. Test whether the signature corner of an arm predicts hallucination or grounding behavior. Use the benchmarks that the sink-intervention literature already uses, such as POPE, CHAIR, and AMBER [13–15]. This paper does not test that last link. It establishes when the signatures form, and in what combinations, which is the measurement that has to exist first.

Reproducibility. A self-validating probe (§2) computes all signatures on a fixed probe batch, from dense logs taken every 100 steps. The Appendix holds the per-seed tables. We will release the code, the probe, the run configurations, the per-run logs, and the training checkpoints on acceptance. We withhold the links for double-blind review.

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Appendix

A. Supporting figures

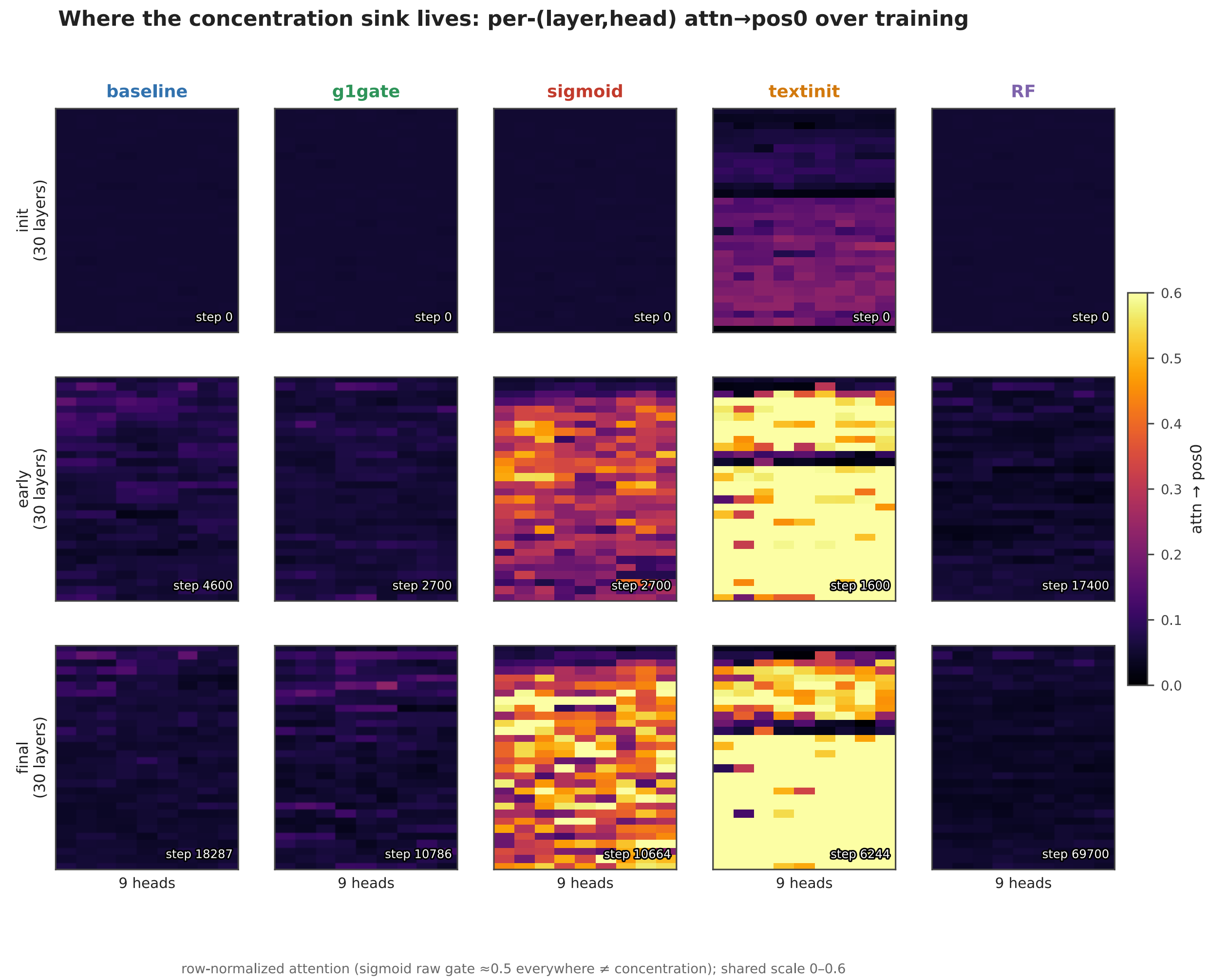

Figure A1: Where concentration lives in the network. Mean attention to pos0 for each of the 270 (layer, query-head) pairs through training, row-normalized; seed 0 for every arm, run to each arm's final checkpoint (174M tokens baseline, 103M g1gate, 102M sigmoid, 60M textinit, 1B RF). baseline and RF stay cold throughout. textinit is hot from initialization. sigmoid lights up a band of mid-network layers.

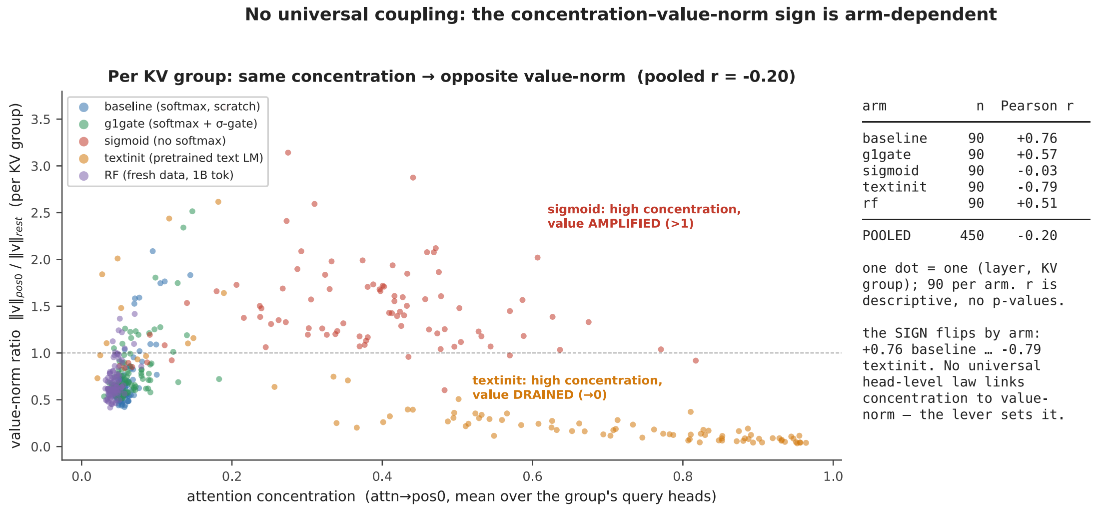

Figure A2: No universal head-level coupling. Concentration against value-norm ratio for each (layer, KV-group) pair at each arm's final checkpoint, seed 0, at n = 90 pairs per arm. Grouping this way avoids triplicating each value-norm observation across the 3 query heads that share it, so these correlations are descriptive and we report no p-values (§3.3). The sign of the correlation flips by arm, from +0.76 in baseline to −0.79 in textinit (Table 4). The pooled cloud is weak, at −0.20, only because arms of opposite sign cancel. Do not read it as an uncorrelated cloud. Most individual arms show a strong |r|.

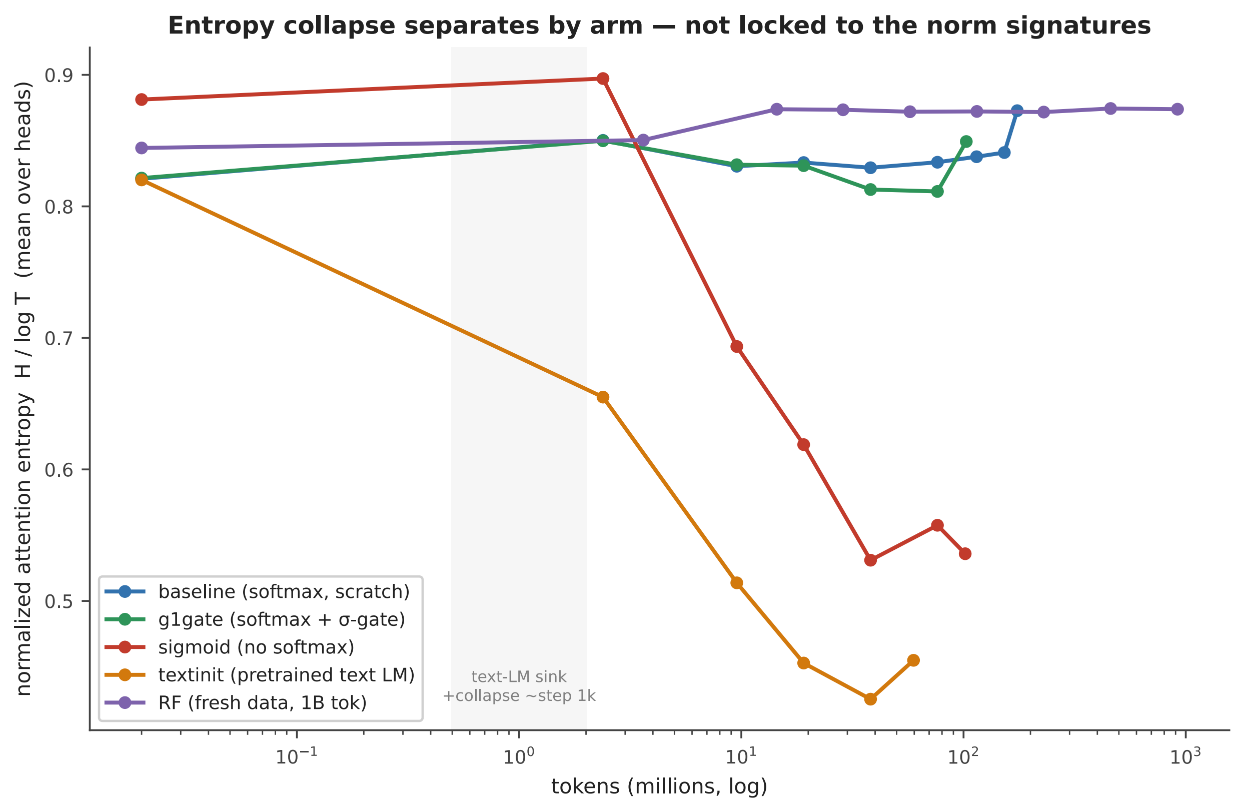

Figure A3: Entropy collapse tracks concentration only. Mean attention entropy through training, seed 0, to each arm's final checkpoint. Only sigmoid and textinit collapse. baseline, g1gate, and RF stay flat. The entropy-collapse correlate from the text-LM literature therefore follows the concentration axis, not the norm axes.

B. Full per-seed signature table

This table reproduces Table 1 and adds the maximum attention→pos0 where we have it. We logged that metric for seeds 1 and 2 only, because the seed-0 archive summary does not include it. We therefore keep it out of the main comparative table.

arm	seed	tokens	$\mathrm{Sink}^{0.2}{}_1$	mean attn→pos0	max attn→pos0	v-ratio	h-ratio
baseline	s0	101M	0.000	0.068	—	0.723	2.16
baseline	s1	100M	0.000	0.062	—	0.687	1.71
g1gate	s0	101M	0.004	0.068	—	0.805	1.67
g1gate	s1	100M	0.011	0.073	~0.21	0.845	2.04
g1gate	s2	100M	0.0037	0.072	~0.22	0.854	2.24
sigmoid	s0	102M	0.830	0.377	—	1.479	1.10
sigmoid	s1	100M	0.756	0.311	—	1.603	1.30
textinit	s0	60M	0.852	0.627	—	0.377	42.5
textinit	s1	60M	0.556	0.235	~0.53	0.634	5.50
textinit	s2	60M	0.578	0.232	~0.59	0.484	12.2

A note on g1gate. Both seeds with max-attention data show one head at about 0.21–0.22 maximum attention→pos0. That single head does clear the strict $\varepsilon = 0.2$ threshold, which is why the arm's $\mathrm{Sink}^{0.2}{}_1$ is small but not exactly zero, and no head comes close to the $\varepsilon = 0.3$ default of [6]. The arm mean meanwhile stays near 0.07 and $\mathrm{Sink}^{0.2}{}_1$ stays far below 0.05. That pattern repeats across seeds.

C. Per-position attention mass (seed 0)

This table gives the mean and max-head attention mass by position at seed 0. It confirms that position 0 is the maximum-mass token in every arm, by mass rather than by argmax count alone. It is the direct check behind the position-0 anchoring defense of §5.

arm	mass@pos0 (mean)	max-head@pos0	argmax-mass position	reading
baseline	0.06	0.17	0	pos0 max; diffuse (no sink)
g1gate	0.07	0.23	0	pos0 max; diffuse
sigmoid	0.30	0.66	0	pos0 max; broad raw-sigmoid mass
textinit	0.63	0.99	0	pos0 max; razor spike (pos1 = 0.009)

C.2 Per-position scan at the remaining seeds, and for RF

We re-dumped per-position attention mass at the other seeds and for RF, and per-position residual and value norms for all three *textinit* seeds. The table gives the position at which each quantity peaks (for value norms, the position at which the norm is smallest).

run	seed	attention-mass argmax	residual-norm peak	value-norm minimum	reading
baseline	s0, s1	0	—	—	pos0 max
g1gate	s0, s1, s2	0	—	—	pos0 max
sigmoid	s0, s1	0	—	—	pos0 max
textinit	s0	0	0	0	all three coincide
textinit	s1	0	1	5	norms move off pos0
textinit	s2	**1**	**13**	**13**	attention separates from coincident norm extrema
RF	s0	1	—	—	mass 0.100 vs 0.083 at pos0; diffuse, not a sink

Two readings follow. Position 0 remains the maximum-mass token in every arm with a randomly initialized decoder, so the pos0 anchoring holds where the central negative result lives. In *textinit* the three signatures need not share a token, which is the positional dissociation reported in §3.4 and the reason the pos0-anchored textinit magnitudes at seeds 1 and 2 understate that arm's peaks.

D. Notes on metric hygiene

We record two corrections that we made during the audit of the numbers in this paper. First, an earlier internal consolidation mixed two metrics in one column, mean and max attention→pos0, and thereby manufactured an apparent seed anomaly in *g1gate*. Re-derived like-for-like, the anomaly disappears, and g1gate becomes the most reproducible arm. Second, an earlier draft claimed that the residual-norm peak of textinit "stays pinned at pos0." We checked that claim against the norm dump. It is wrong. The ‖h‖ peak sits at pos0, pos1 or pos13, depending on the seed (Appendix C.2). That is the positional dissociation reported in §3.4, which states the corrected form.